

USRP Studies

Week 1: Basic Asymptotics

Date

Overview

This lecture went over the bread and butter of runtime analysis.

2a. Big O and friends

Definition 1.1. (*Big-Oh*) $f(x) = O(g(x))$ iff $\exists C, x_0 > 0$ s.t. $|f(x)| \leq Cg(x) \forall x \geq x_0$

Remark 1.1. In TCS, runtime is inherently nonnegative, so abs is often omitted.

Remark 1.2. $O(g(x))$ denotes an anonymous function that grows no faster than $g(x)$.

Example: $\sum_{i=1}^n i = \frac{1}{2}n^2 + O(n)$.

Definition 1.2. (*Big-Omega*) $f(x) = \Omega(g(x))$ iff $\exists c, x_0 > 0$ s.t. $f(x) \geq cg(x) \forall x \geq x_0$

Definition 1.3. (*Theta*) $f(x) = \theta(g(x))$ iff $f(x) = O(g(x))$ & $f(x) = \Omega(g(x))$

Definition 1.4. (*Little-O and Little-Omega*) o is strictly less than, ω is strictly more than

Intuition: Big-Oh bounds f above, Big-Omega bounds f below, Theta bounds f in between.

Definition 1.5. Some notation

- $f(x) \stackrel{\text{nota}}{=} \text{poly}(g(x)) \equiv g(x)^{O(1)}$
- $f(x) = \tilde{O}(g(x)) \equiv f(x) = g(x) \cdot \text{poly} \log g(x)$
- $f(x) = \tilde{\Omega}(g(x)) \equiv f(x) \geq \frac{g(x)}{\text{poly} \log g(x)}$
- $f(x) \sim g(x) \text{ (as } x \rightarrow \infty) \equiv \frac{f(x)}{g(x)} \rightarrow 1$

Definition 1.6. $g(n)$ is in *standard form* if it is a product of...

1. constants
2. constant powers of $\ln n$
3. constant powers of n
4. exponential functions
5. n^{cn} , c constant

Theorem 1.7. (*Limit theorem for little-oh*) $(1) \ll (2) \ll (3) \ll (4) \ll (5)$

Theorem 1.8. (*Stirling's approximation*) $n! \sim \sqrt{2\pi n} \cdot n^n \cdot e^{-n}$

2b. Harmonic Numbers Asymptotics

In this part, we do an analysis of the n th harmonic number. This number shows up all the time in TCS (e.g. Coupon collector's problem)

Definition 1.9. $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$

We can try to naively bound H_n this way:

$$1 + \frac{1}{2} + 2 \cdot \frac{1}{4} + 4 \cdot \frac{1}{8} + \dots \leq H_n \leq 1 + 2 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} + 8 \cdot \frac{1}{8} + \dots$$

$$\frac{1}{2} \lfloor \log_2 n \rfloor + 1 \leq H_n \leq \lfloor \log_2 n \rfloor + 1$$

$$\therefore H_n = \theta(\log n)$$

Theorem 1.10. $H_n \sim \ln n$

Proof. We consider the function $f(t) = \frac{1}{t}$. The area of the sum of rectangles $f(t) \cdot 1$ must be less than or equal to $1 + \int_1^n \frac{1}{t} dt = 1 + \ln n$. Now we shift the rectangles in our picture by 1. So $H_n \geq \int_1^{n+1} \frac{1}{t} dt = \ln(n+1)$. Hence, $\ln(n+1) \leq H_n \leq \ln(n) + 1$, and $H_n \sim \ln n$. $\square$

Theorem 1.11. $H_n = \ln n + \gamma - O(\frac{1}{n})$, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant.

Proof. An application of Taylor Series. $\square$

2c. Asymptotic Tricks

We can consider asymptotics to analyze various common expressions.

Important Taylor Series approximations for small values

1. $e^x \approx 1 + x$
2. $\frac{1}{1-\varepsilon} \approx 1 + \varepsilon$
3. By (1) and (2), $e^\varepsilon \approx \frac{1}{1-\varepsilon}$
4. $\sqrt{1+\varepsilon} \approx 1 + \frac{1}{2}\varepsilon$
4. $\sqrt{e^\varepsilon} = e^{\varepsilon/2} \approx 1 + \frac{\varepsilon}{2}$

Analyzing how an inverse behaves

Suppose we have $f(x) = y$ and want to find $f^{-1}(y)$. We can take logs of both sides, and keep only the dominant term to get an approximation for x .

Minimizing / Maximizing with a parameter

When our objective function has a term that adds, and a term that subtracts to the parameter, the optimum is attained, up to a constant factor, when they are roughly equal to each other.

Appendix

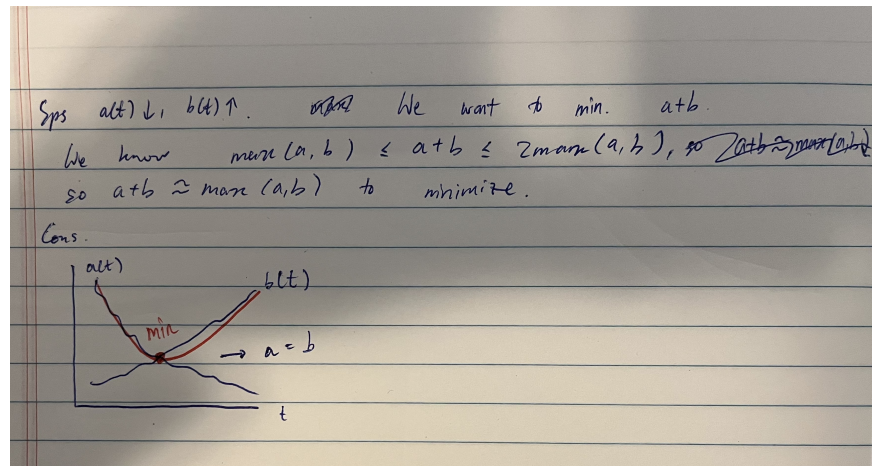


Figure 1: A heuristic on minimizing $a(t) + b(t)$ by balancing the two terms.